# Polytrauma for team communication



## Section I: Scenario Demographics

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|---|---|
| Scenario Title: | Basic polytrauma for trauma team communication |
| Date of Development: | (01/05/2018) |
| Target Learning Group: | ☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups |

## Section II: Scenario Developers


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| Scenario Developer(s): | Dr. Chris Heyd |
| Affiliations/Institution(s): | McMaster University |
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## Section III: Curriculum Integration

| Learning Goals & Objectives | |
|---|---|
| **Educational Goal:** | To challenge learners' teamwork and communication skills through the stabilization of an unstable trauma patient |
| **CRM Objectives:** | 1. To practice effective leadership and followership in a clinical team<br>2. To use best practices in team communication, including closed loop communication and periodic summaries<br>3. To perform critical procedures during a resuscitation without losing situational awareness |
| **Medical Objectives:** | 1. To recognize respiratory failure in a blunt trauma patient and initiate simultaneous treatment and diagnostics<br>2. To safely intubate an unstable trauma patient<br>3. To identify and expediently treat a large hemothorax<br>4. To identify and treat decompensated hemorrhagic shock |

| Case Summary: Brief Summary of Case Progression and Major Events |
|---|
| A 64-year old man is involved in a high-speed car crash. The trauma team is activated and he is brought directly to the ED. On arrival, he is hypoxic, tachycardic and altered. CXR reveals multiple rib fractures with a right-sided hemopneumothorax.<br><br>The team leader will need to effectively communicate with the team to ensure the tasks of intubation, chest tube placement and blood product administration are performed in a safe and quickly. The patient will stabilize after these treatments.<br><br>Members of the trauma team will have a staggered entry into the room. The team leader will need to balance communication with the new team members and the urgent interventions needed by the patient. |



| References |
| --- |
| Marx JA, Hockberger RS, Walls RM, & Adams J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby. |
| Gillman LM, Widder S, Blaivas M & Karakitsos D. (2016). *Trauma team dynamics: A trauma crisis resource management manual*. Switzerland: Springer. |

## Section IV: Scenario Script

| A. Scenario Cast & Realism | | | |
| --- | --- | --- | --- |
| Patient: | ☒ Computerized Mannequin | Realism: | ☒ Conceptual |
| | ☐ Mannequin | | ☐ Physical |
| | ☐ Standardized Patient | *Select most* | ☐ Emotional/Experiential |
| | ☐ Hybrid | *important* | ☒ Other: Interprofessional |
| | ☐ Task Trainer | *dimension(s)* | ☐ N/A |
| **Confederates** | **Brief Description of Role** | | |
| Bedside RN | To prompt team to clinical status, as required | | |
| | | | |
| | | | |

| B. Required Monitors | | |
| --- | --- | --- |
| ☒ EKG Leads/Wires | ☐ Temperature Probe | ☐ Central Venous Line |
| ☒ NIBP Cuff | ☐ Defibrillator Pads | ☒ Capnography |
| ☒ Pulse Oximeter | ☐ Arterial Line | ☐ Other: |

| C. Required Equipment | | |
| --- | --- | --- |
| ☒ Gloves | ☒ Nasal Prongs | ☒ Scalpel |
| ☒ Stethoscope | ☒ Venturi Mask | ☒ Tube Thoracostomy Kit |
| ☐ Defibrillator | ☒ Non-Rebreather Mask | ☐ Cricothyroidotomy Kit |
| ☒ IV Bags/Lines | ☒ Bag Valve Mask | ☐ Thoracotomy Kit |
| ☒ IV Push Medications | ☒ Laryngoscope | ☐ Central Line Kit |
| ☐ PO Tabs | ☒ Video Assisted Laryngoscope | ☐ Arterial Line Kit |
| ☒ Blood Products | ☒ ET Tubes | ☒ Other: C-Collar |
| ☐ Intraosseous Set-up | ☐ LMA | ☐ Other: |

| D. Moulage |
| --- |
| Bruising to the right side of the thorax |

| E. Approximate Timing | | | | | |
| --- | --- | --- | --- | --- | --- |
| Set-Up: | Variable for in situ | Scenario: | 10 min | Debriefing: | 10 min |



## Section V: Patient Data and Baseline State

### A. Clinical Vignette: To Read Aloud at Beginning of Case

*To be read aloud by simulation facilitator at start of case*: "You are working as an Emergency physician at a tertiary care trauma center and have been called overhead to your trauma bay. A paramedic team has just arrived with a 64-year old trauma patient. He was involved in a highway speed head-on MVC. He was restrained and air bags deployed. He was the driver and the other drive died on scene. There were no other passengers. EMS extricated the patient easily. They have placed one IV line and started running normal saline. He has been placed on a non-rebreather mask but has remained tachycardic, hypoxic and altered. GCS has been consistently 14. The trauma team was activated based on injury mechanism but so far only the orthopedic resident has arrived at the bedside."

### B. Patient Profile and History

| | | |
|---|---|---|
| Patient Name: Paulo Matos | Age: 64 | Weight: 80 kg |
| Gender: ☒ M ☐ F | Code Status: Unknown | |
| Chief Complaint: Blunt polytrauma | | |
| History of Presenting Illness: High-speed MVC, head-on collision, restrained + airbags | | |

| | | | |
|---|---|---|---|
| Past Medical History: | Hypertension | Medications: | Hydrochlorothiazide |
| | Dyslipidemia | | Ramipril |
| | | | Atorvastatin |

Allergies: None

Social History: Retired steel factory foreman

Family History: Unremarkable

| | | | | |
|---|---|---|---|---|
| Review of Systems: | CNS: | Amnestic to event, mild confusion | | |
| | HEENT: | Nil | | |
| | CVS: | Sharp chest pain, pleuritic | | |
| | RESP: | Short of breath | | |
| | GI: | Nil | | |
| | GU: | Nil | | |
| | MSK: | Nil | INT: | Nil |

### C. Baseline Simulator State and Physical Exam

☐ No Monitor Display ☒ Monitor On, no data displayed ☐ Monitor on Standard Display

| | | | |
|---|---|---|---|
| HR: 120/min | BP: 100/65 | RR: 28/min | $O_2$SAT: 92% (NRB) |
| Rhythm: Sinus tach | T: 36.5°C | Glucose: 6.5 mmol/L | GCS: 14 (E4 V4 M6) |

General Status: Tachycardic, tachypneic, hypoxic, confused

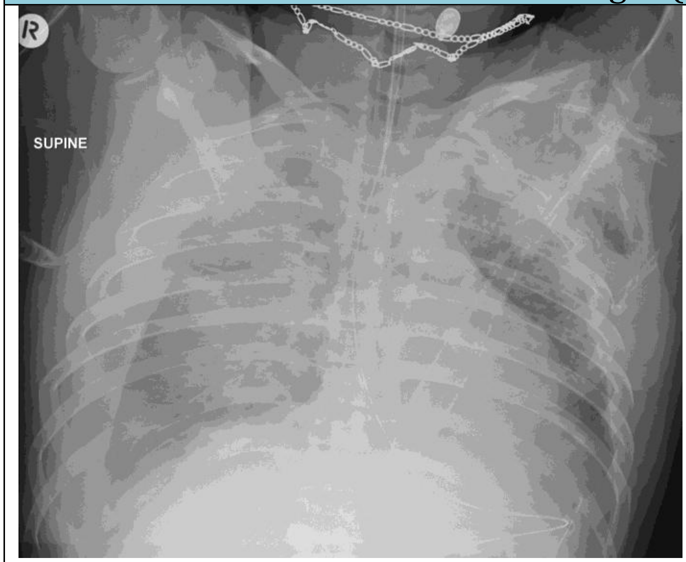
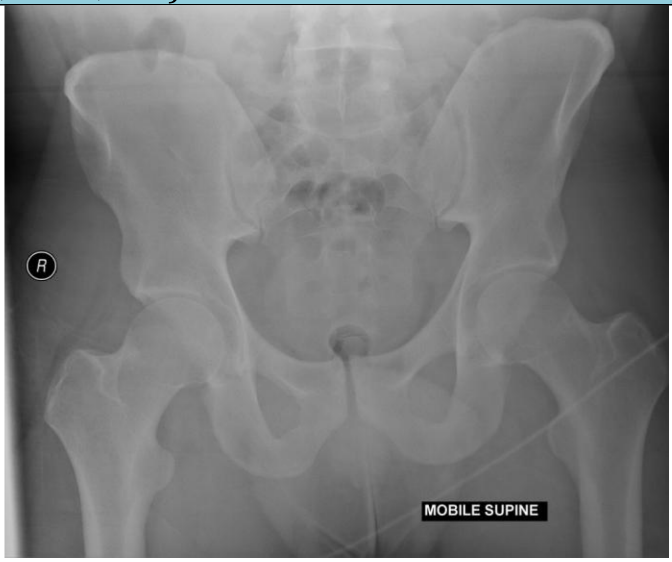
| | | | |
|---|---|---|---|
| CNS: | Awake, alert, confused, amnestic. PERL, moving all limbs, complaining of chest pain | | |
| HEENT: | Unremarkable | | |
| CVS: | Tachycardic, heart sounds normal | | |
| RESP: | Tachypneic, decreased breath sounds on Right side, crepitus | | |
| ABDO: | Soft, non-tender, non-distended | | |
| GU: | Unremarkable | | |
| MSK: | Right chest tenderness+crepitus<br>No long bone deformity, pelvis stable | SKIN: | Right chest bruising |

## Section VI: Scenario Progression

| Scenario States, Modifiers and Triggers | | | |
|---|---|---|---|
| Patient State | Patient Status | Learner Actions, Modifiers & Triggers to Move to Next State | |
| **1. Baseline State**<br>Rhythm: Sinus tach<br>HR: 120/min<br>BP: 110/65<br>RR: 28/min<br>$O_2$SAT: 92% (NRB)<br>T: 36.6°C | Confused, complains of chest pain | <u>Learner Actions</u><br>-☐ Place on monitors<br>-☐ Listen to EMS<br>-☐ Place $2^{nd}$ IV line + BW<br>-☐ Start crystalloid<br>-☐ Call for blood products<br>-☐ Primary survey<br>-☐ FAST<br>-☐ Provide analgesia<br>-☐ Call for portable CXR, PXR<br>-☐ Cap glucose | <u>Modifiers</u><br>*Changes to patient condition based on learner action*<br>***- EMS handover finishes → Charting RN enters***<br><br><u>Triggers</u><br>*For progression to next state*<br>- All actions complete or **5 min** into case → **2. Hypoxia** |
| **2. Hypoxia**<br>Rhythm: Sinus tach<br>HR: 125/min<br>BP: 110/65<br>RR: 28/min<br>$O_2$SAT: 86% (NRB) | Becoming agitated | <u>Learner Actions</u><br>-☐ Communicate situation to anesthesia<br>-☐ Prepare for intubation<br>-☐ Prepare for chest tube | <u>Modifiers</u><br>***- Start of state → Anesthesia enters***<br>- Attempt chest tube placement → patient becomes agitated<br><br><u>Triggers</u><br>- Intubation → **3. Hypotension** |
| **3. Hypotension**<br>Rhythm: Sinus tach<br>HR: 110/min<br>BP: 75/50<br>RR: 12/min<br>$O_2$SAT: 90% (ETT) | Intubated | <u>Learner Actions</u><br>-☐ Communicate situation to TTL<br>-☐ Needle or finger thoracostomy<br>-☐ Place chest tube<br>-☐ Blood products<br>-☐ TXA | <u>Modifiers</u><br>***- Start of state → TTL enters***<br>- Chest tube placed → $O_2$SAT=96%<br><br><u>Triggers</u><br>- Chest tube placed and blood products started → **4. Resolution** |
| **4. Resolution**<br>Rhythm: Sinus tach<br>HR: 110/min<br>BP: 90/65<br>RR: 12/min<br>$O_2$SAT: 96% (ETT) | Intubated | <u>Learner Actions</u><br>-☐ Post intubation sedation<br>-☐ Call CT<br>-☐ Call ICU | <u>Modifiers</u><br><br><u>Triggers</u><br>- Call CT → **End Case** |

Section VII: Supporting Documents, Laboratory Results, & Multimedia

| Laboratory Results |
|---|
| Not given during case |

| Images (ECGs, CXRs, etc.) | |
|---|---|
|   Case courtesy of Dr Andrew Dixon, https://radiopaedia.org/<br>From the case https://radiopaedia.org/cases/31573 rID: 31573 |   Case courtesy of Dr Craig Hacking, https://radiopaedia.org/<br>From the case https://radiopaedia.org/cases/41751 rID: 41751 |

| Ultrasound Video Files (if applicable) | |
|---|---|
| Negative FAST | Chest ultrasound confirms right-sided hemopneumothorax |

## Section VIII: Debriefing Guide

### General Debriefing Plan

| | | | |
|---|---|---|---|
| ☐ Individual | ☒ Group | ☐ With Video | ☒ Without Video |

### Learning Goals & Objectives

| | |
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### Sample Questions for Debriefing

**CRM**

1. What communication challenges were present during this patient encounter? What tools did you use to overcome them?
2. What steps did you use to prepare the team for the specific tasks (intubation, chest tube) that were required for this patient?
3. What is your strategy for maintaining awareness of the whole patient while specific tasks are being performed?

**Medical Expert**

1. Describe your differential diagnosis for respiratory failure in a trauma patient? How did you work through this in this patient?
2. What are the indications for intubation in a trauma patient? What was the indication in this patient?
3. Outline an approach to intubation of the trauma patient.
4. What is the clinical evidence for hemorrhagic shock? What is the difference between compensated and decompensated shock?

### Key Moments

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|---|
| Conducting a primary survey |
| Preparing the team for intubation and chest tube placement |
| Recognizing hemorrhagic shock and ordering blood products |